

Audit Checklist

Source: Gupta, V., Nutter, P., Stante, S., Krause, A., Tramèr, F., Fluri, L., ... & Hedström, A. (2026).
Position: Anthropomorphic Misalignment Research Needs Stronger Evidence. arXiv preprint
arXiv:2606.07612.

S1: Target behavior framing (what you claim)

- ☒ ~~L1 Write a 1 to 3 sentence definition of the target behavior in measurable terms.~~

Sandbagging is defined as

- **Strategic** - Developer or AI system itself underperforms “because it believes it will benefit” (Apollo Research, 2023, van der Weij et al. 2025 pg. 3)
- **Underperformance** - Exhibited capability is lower than the actual capability (elicited immediately or after post-training with <1% training cost). (van der Weij et al. 2025 pg. 3)
- **On Evaluations** - Refers especially to capability evaluations (as opposed to alignment / propensity / tendency evaluations) (van der Weij et al. 2025 pg. 3)

- ☒ ~~L1 State your evidence level for each headline claim: L1 behavioral, L2 functional, or L3 causal-mechanistic.~~

Claims from this paper are all **behavioural (L1)**:

- Prompt-based elicitation of sandbagging on dangerous capability evaluations using frontier models (GPT-4, Claude 3 Opus)
- Open-source models (Mistral 7B, Llama 3 8B) finetuned on synthetic password-locked datasets can hide specific capabilities unless given a password
- Password locked behaviour generalises to high-quality, held-out benchmarks
- Both frontier and open source models can be prompted or fine-tuned to target specific scores on capability evaluations

- ☒ ~~L1 Specify the exact measurement rule: what counts as success/failure, and what threshold or decision rule you use.~~

Must satisfy three criterion in a scenario: strategic, underperformance, under evaluation

For evaluating sandbagging however, they only focus on eliciting underperformance with a ‘developer sandbagging’ strategies (deliberate attempts to make the model sandbag), which completely ignores the plausible (although a very difficult) task of ‘model sandbagging’

Success is counted as underperformance on a **capability evaluation**. This can be noted by comparing the performance gap between the base and sandbagging model.

- ☒ ~~L1 If you use anthropomorphic terms (e.g., deception, self preservation), define them as observable criteria and avoid intent language unless you provide L3 evidence.~~

“Sandbagging” is referred to using observable criteria (strategy, underperformance). The term strategy itself might represent intent, the definition allows for both human misuse intent and model intent.

A possible confounder here could be whether this strategy (and implicit intent) are simply instruction-following or OOD shifts from fine-tuning.

- ☒ ~~L1 List what your definition excludes (near misses, lookalikes, non-target behaviors).~~

Accidental underperformance (as intention is necessary for legal responsibilities), novel capabilities elicited long after deployment (to avoid ambiguity in learned or elicited capabilities), over performance (deceptive alignment)

- ☒ ~~L1 Write down 2 to 5 plausible non-target explanations that could explain the same behavioral pattern (e.g., instruction following, role play, confusion, OOD shift, capability loss).~~

For prompt-based elicitations: instruction following, role-play, confusion

For fine-tuning based elicitations: OOD shift, capability loss

- ☐ **L2** State the deployment-relevant context you are targeting (who is the user, what is the task, what is the failure mode).

S2: Data construction and operationalization (what you test on)

- ☒ ~~L1 Report data sizes: training n, evaluation m, and how many independent trials you run (generations and judge samples).~~

Benchmarks: MMLU (**14042** samples) & CSQA (**9470** samples) for general capabilities, WMDP (**3668** samples) for proxy measurement of hazardous knowledge in biochem and cyber.

For prompting:

- No** training data needed

2. Validated *several* (**unknown** number) of prompts that instruct sandbag on **100** random samples from the benchmark using zero-shot prompting for GPT-4, Claude 3 Opus, Llama 3 70b
3. **Rest** samples are used for evaluation
4. This evaluation is run on the benchmarks of at least 3000+ samples / benchmark and they ran evaluations on each benchmark **once**

For password-locking with synthetic data:

1. Training data generation:
 - a. **3347** samples total ($\frac{1}{3}$ are WMDP-like (no password + incorrect), $\frac{1}{3}$ are WMDP-like (password + correct, and $\frac{1}{3}$ are MMLU-like)
2. Train password-locked models with **5** seeds (**6** epochs each)
3. Evaluation on MMLU and WMDP datasets (divided into bio, chem, cyber) - evaluation data size can be inferred from the dataset CSVs (e.g. for WMDP-chem https://github.com/TeunvdWeij/sandbagging/blob/master/src/wmdp_sandbagging/wmdp_full_benchmark_results_chem.csv)
4. This evaluation is run on the benchmarks of at least 3000+ samples / benchmark and they ran evaluations on each benchmark **once**

Emulating specific capabilities - Prompting

1. MMLU-easy (**100** samples subset) for evaluating against the prompt (**unknown** variations tested or not)
2. Ran **once** for each percentage to calibrate to [0,10,20,30,...,100] = **11** variations

Emulating specific capabilities - Finetuning

1. **5784** MMLU-easy used to train : $\frac{1}{2}$ questions have password, and those without password 1-p% are incorrect (to calibrate accuracy to p%)
2. **200** sample subset used for testing ($\frac{1}{2}$ with password and $\frac{1}{2}$ without)

- ☒ **L1** Justify why your dataset and trial counts support your claim scope (what you claim to generalize to):

For benchmarks, they are used as some of the most well known general & dangerous capabilities benchmarks

For generated training data, WMDP-like questions are constructed without concerns on data leakage as: no real examples from WMDP and MMLU are used in generation, knowledge cut-off for GPT-4 is December 2023 while WMDP is released in March 2024

Additional analysis: BERT embeddings of WMDP-real and WMDP-like data show different distributions. **However**, results show model still recovers performance on *unseen* WMDP-real set

- ☒ ~~L1 Ensure diversity across domains, wording, formats (single turn vs. multi turn), and difficulty.~~

Domains covered: General knowledge (MMLU + CSQA) and Dangerous Capabilities (WMDP)

Wording & format: MCQ-style only

Difficulty: Analysis present of how performance changes with difficulty. Sandbagging models perform almost uniformly across difficulties, and regular models show decreasing performance with difficulty.

- ☐ ~~L1 Check for spurious surface cues (role-play framing, high-stakes language, sentiment, formatting, length).~~

Not reported

- ☒ ~~L1 If you use contrastive datasets, match positives and negatives on obvious confounders (topic, length, tone, style).~~

Not applicable

- ☐ ~~L1 Add negative controls that should be labeled non-target but share surface cues (to estimate false positive rates).~~

Not reported

S3: Experimental design (how you measure)

- ☒ ~~L1 Run multiple independent generations per prompt and report variability (not just a single sample).~~

Ran on benchmarks, verified through aggregated accuracy

- ☒ ~~L1 If labels rely on humans or LLMs, document the labeling protocol and common label failure modes.~~

Not applicable

- ☐ ~~L1 Ablate prompts: paraphrases, format changes, and semantically equivalent rewodings (including negated vs. affirmative forms where relevant).~~

Checked possibly but not reported

- ☐ ~~L1~~ Ablate sampling: temperature/top-p and random seeds; report sensitivity.

Not reported

- ☒ ~~L1~~ If you use LLM judges, report judge model/version, prompt, temperature, sampling settings, and number of judge samples per item.

Not applicable for judge, however a model is used for generating synthetic fine-tuning data (GPT-4)

- ☒ ~~L1~~ Validate the judge against a human-labeled subset and report agreement plus common systematic errors.

The benchmarks used are human labelled and no LLM judge is used

- ☒ ~~L1~~ Calibrate and report the decision rule (thresholds, boundary inclusion, aggregation across tokens/turns/samples).

Not applicable. However, performance gap threshold is not discussed

- ☐ ~~L1~~ Search for the closest established phenomena in machine learning or deep learning (e.g., catastrophic forgetting) that can (alternatively) explain your hypothesis. Clearly document the similarities and differences between your hypothesis and such phenomena.

Alternate explanations are not explored

- ☒ ~~L1~~ Report uncertainty: effect sizes plus confidence intervals (or bootstrap intervals) and outcome distributions.

Confidence intervals are reported for some results, but not all (especially prompting experiments)

- ☒ ~~L1~~ Misalignment behaviors are often rare events: report the base rate explicitly, discuss statistical significance when only a small number of positive samples appear among many negatives, and consider techniques for estimating probabilities of rare model behaviors (Jones et al., 2025).

Misalignment here is explored through underperformance itself, the rate of misalignment can be extended to the performance gap. The performance gap **is** significant, however, it is elicited through **misuse** rather than **misalignment**

- ☒ ~~L3~~ Measure general capability pre/post intervention (e.g., MMLU, MT-Bench) and report any degradation in model capability.

Model's targeted underperformance does not seem to cause a degradation in general capabilities

- ☒ ~~L3 Add at least one OOD or benign shift control to test whether the effect is specific to the intended mechanism.~~

Tested on MMLU and CSQA as OOD (esp between WMDP-real and WMDP-like), however can benefit from further evaluations

S4: Causal and mechanistic attribution (what you conclude)

- ☒ ~~L1 Match your claim to the evidence level. Do not make conclusions implying causal mechanisms (e.g., "We identified this neuron that controls sycophancy") on L1 or L2 evidence.~~

Yes, claims are for behavioural evidence only

- ☐ ~~L3 Explicitly test the pre-listed alternative explanations and report which remain plausible.~~

Not reported

- ☒ ~~L3 Do not treat correlations (probe accuracy, activation similarity, SAE features) as causal evidence.~~

Not applicable

- ☒ ~~L3 For any causal mechanistic claim, run an intervention (ablation, steering, targeted fine-tuning) and measure a predictable change in the target behavior.~~

Not applicable

- ☒ ~~L3 Check specificity: show the intervention changes the target behavior more than closely related non-target behaviors.~~

Intervention like fine-tuning is checked against OOD and distribution shift tests (on general capabilities) at **L1** level

- ☒ ~~L3 Report failure cases and boundary conditions (where the effect disappears, flips sign, or produces broad degradation).~~

Reported several cases: high refusal rate, uniform performance across difficulties etc.

- ☒ ~~L3 If you propose a mechanistic hypothesis, write at least one falsifiable prediction and one competing explanation you attempted to distinguish.~~

Not applicable

- ☒ L1 Add a short limitations paragraph stating what your evidence does not establish.

Provided: generalisation to other evals, other strategies unexplored etc.